

Digital Mapping Cameras

IRS Chapter Four

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Roadmap

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4.2 Fundamentals of the Aerial Photograph

4.3 Geometry of Vertical Aerial Photographs

4.4 Digital Aerial Cameras

4.5 Digital Scanning of Analog Images

4.6 Spectral Sensitivity

4.7 Band Combinations in Optical Imagery

4.8 Coverage by Multiple Photographs

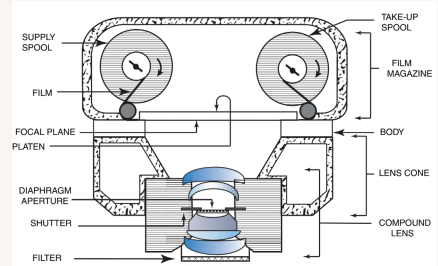
4.9 Photogrammetry

4.10 Acquiring Aerial Photography

Resources

Digital Mapping Cameras

- ▶ Cameras: **oldest remote sensing tool**
- ▶ Continuity in purpose despite **technological change**
- ▶ High **positional accuracy** and **spatial detail**
- ▶ Aerial photography used by:
 - ▶ Governments
 - ▶ Private businesses
 - ▶ Federal agencies
- ▶ **Versatility** and **archival value** since 1930s



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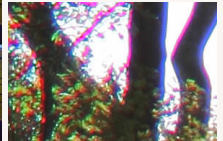
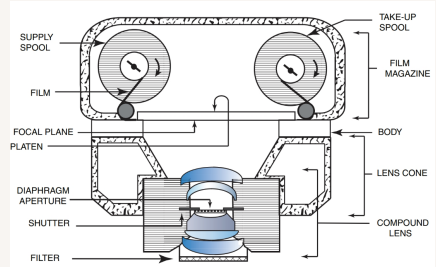
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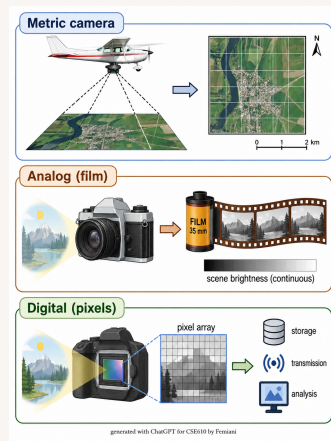
Fundamentals of Aerial Photographs

- ▶ Key components of aerial cameras:
 - ▶ **Lens:** Gathers and focuses light
 - ▶ **Light-sensitive surface:** Records the image
 - ▶ **Shutter:** Controls light entry
 - ▶ **Camera body:** Holds components
- ▶ **Film magazine and drive mechanism** unique to aerial cameras
- ▶ **Metric cameras:** High positional accuracy
 - ▶ Calibration ensures accurate measurements and corrects lens distortions.
 - ▶ Avoid optical aberrations such as distortion and chromatic aberration



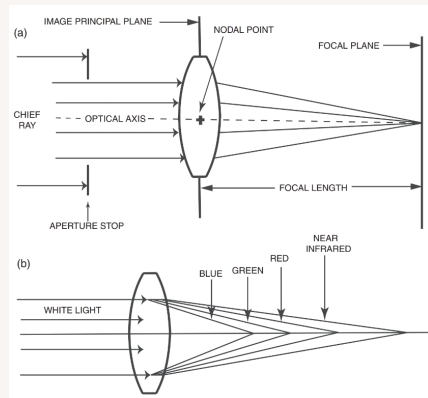
Key Terminology: Aerial Photography

- ▶ **Metric (Cartographic) Cameras:**
 - ▶ High-quality imagery with **positional fidelity**
 - ▶ Optimized for **mapping** applications
- ▶ **Analog Photography:**
 - ▶ **Film-based** images
 - ▶ Brightness proportional to scene brightness
- ▶ **Digital Photography:**
 - ▶ Images as **pixel arrays**
 - ▶ Easier **storage, transmission, and analysis**

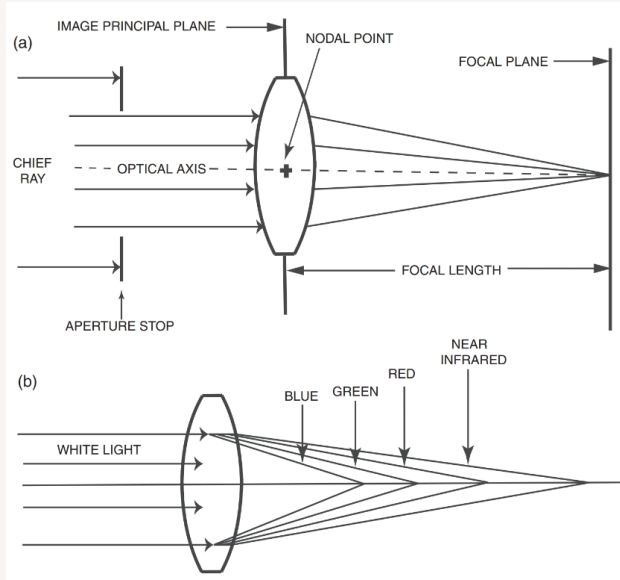


Lens vs. Pinhole Camera

- ▶ **Pinhole Camera:** Small hole lets light in
 - ▶ **No lens** or adjustable focus
 - ▶ Needs long focal length and exposure
 - ▶ Limited light, so images can be dim
- ▶ **Lens Camera:** Uses glass to focus light
 - ▶ **Controls focus** for sharpness
 - ▶ Allows for **faster exposure** and more light
- ▶ **Lens Advantage:** More light, greater control over exposure

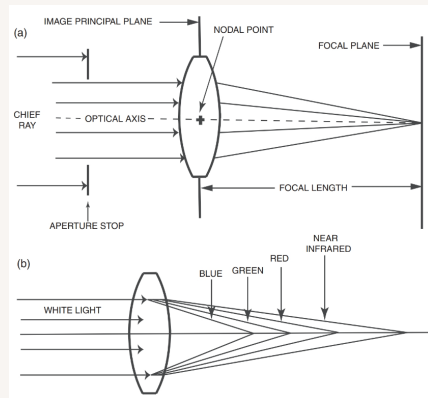


The Lens in Aerial Photography



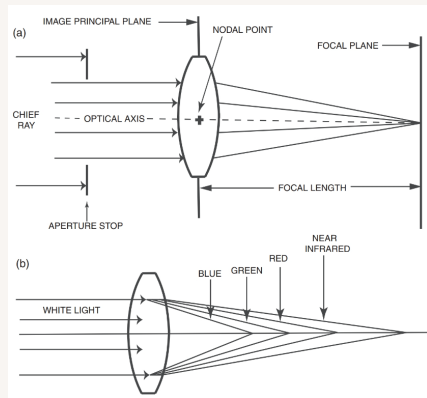
The Lens in Aerial Photography

- ▶ **Lens:** Gathers and focuses light
- ▶ **Optical Axis:** Central path of light through lens
- ▶ **Spherical Aberration:** Image distortion due to lens imperfections
- ▶ **Chromatic Aberration:** Different wavelengths focus at varying points
- ▶ **High-quality lenses:** Correct for these aberrations



Compound Lenses and Optical Properties

- ▶ **Compound Lens:** Multiple lenses combined to minimize distortions
- ▶ **Optical Axis:** Central line through the lens that light follows
- ▶ **Image Principal Plane:** Theoretical center of refraction in a lens
- ▶ **Nodal Point:** Point where the optical axis and image principal plane meet
- ▶ **Focal Point:** Where parallel light rays converge to form the image



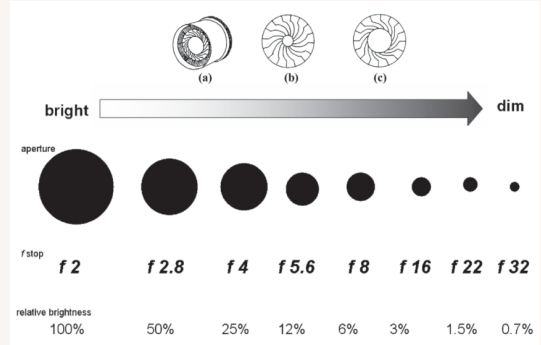
Focal Length and Aperture

- ▶ **Focal Length:** Distance from lens to focal point
- ▶ **Aperture:** Adjustable opening controlling light entry
- ▶ **f-number (Relative Aperture):**

- ▶ Defined as:

$$f = \frac{\text{Focal length}}{\text{Aperture size}}$$

- ▶ Controls **brightness** of the image
- ▶ Large f-number = **small aperture**
- ▶ Small f-number = **large aperture**



F-stop

Each **f-stop** each represents a halving or doubling of light entering the camera. For example, $f/2.8$ lets in twice as much light as $f/4$.

Vignetting and Shutter Mechanisms

- ▶ **Vignetting:** Dark edges in images due to light falloff
 - ▶ More pronounced at the edges of wide-field lenses
- ▶ **Antivignetting Filters:** Compensate for brightness falloff
 - ▶ Darker at center, clearer at edges
 - ▶ Corrects uneven light distribution



Image Motion Compensation (IMC)

- ▶ Prevents image blur from **aircraft movement**
- ▶ Adjusts film or sensor speed to match **aircraft velocity**
- ▶ **Analog cameras:** Film moved mechanically
- ▶ **Digital cameras:** Compensation done electronically
- ▶ Enables clear imagery at **low altitudes** and **high speeds**



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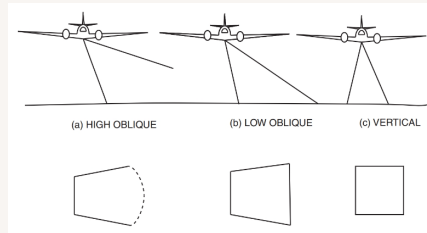
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Geometry of Vertical Aerial Photographs

- ▶ **Vertical photographs:** Camera points directly down at the ground
- ▶ **Oblique photographs:** Camera angled sideways
 - ▶ **High oblique:** Shows horizon
 - ▶ **Low oblique:** Does not show horizon
- ▶ **Practical advantages** of vertical photos:
 - ▶ **Consistent scale**
 - ▶ Suitable for accurate **measurement**
- ▶ Most aerial photos have a slight tilt

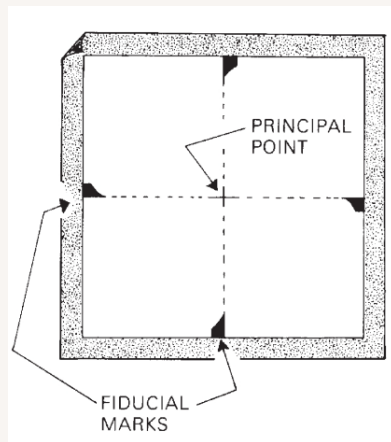






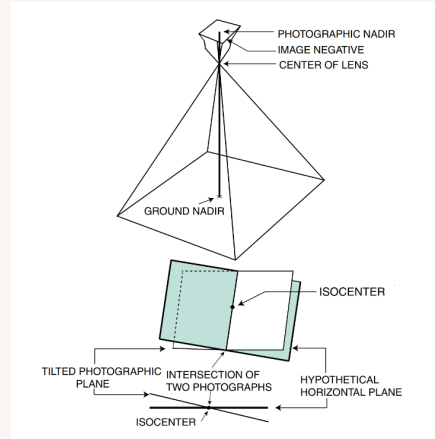
Fiducial Marks and Key Geometric Points

- ▶ Mainly for analog images
- ▶ **Fiducial Marks:** Reference points for accurate measurement
 - ▶ Recorded on photographs during exposure
 - ▶ Typically four or eight marks
- ▶ **Principal Point:** Intersection of the **optical axis** and the **focal plane**



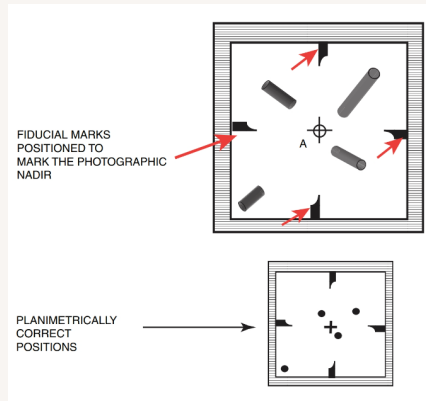
Fiducial Marks and Key Geometric Points

- ▶ **Ground Nadir:** Point on the ground directly below the camera
- ▶ **Photographic Nadir:** Where the vertical line from the ground nadir intersects the photograph
- ▶ **Isocenter:** Focus of tilt, where image tilt occurs



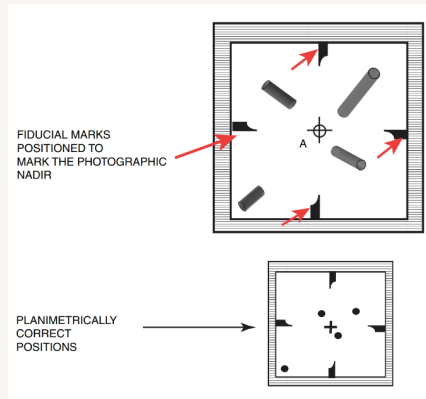
Positional Errors in Vertical Aerial Photographs

- ▶ **Optical Distortions:** Caused by lens issues or camera malfunction
 - ▶ Minor in modern photography with high-quality equipment
- ▶ **Tilt:** Displacement of the focal plane from horizontal
 - ▶ Upper areas appear smaller, lower areas larger
 - ▶ Systematic errors can arise if measurements are far from the principal point
- ▶ **Relief Displacement:** Objects “lean” outward from the nadir
 - ▶ More displacement for taller objects and those farther from the nadir
 - ▶ Caused by central perspective, inherent to aerial photographs



Relief Displacement in Vertical Photographs

- ▶ **Relief Displacement:** Objects appear to lean outward from the nadir
 - ▶ Increases with object height and distance from nadir
- ▶ **Factors affecting displacement:**
 - ▶ **Height of the object**
 - ▶ **Distance from the nadir**
- ▶ **Measurement impact:**
 - ▶ Causes positional errors in uncorrected photographs
 - ▶ More significant in uneven terrain
- ▶ **Radial displacement:** Outward from the nadir toward the edges



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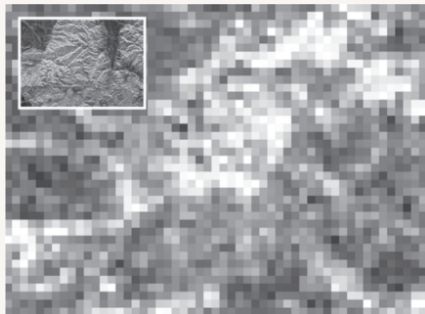
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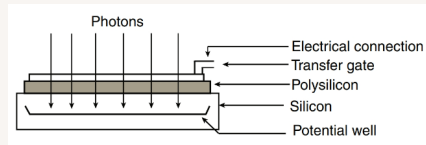
Digital Aerial Cameras

- ▶ **Pixels:** Small regions of the Earth's surface captured in brightness values
 - ▶ Each pixel stores digital brightness values
- ▶ **Matrix of pixels** forms the complete image
- ▶ **Analog vs. Digital:**
 - ▶ **Analog:** Uses film coated with photosensitive chemicals
 - ▶ **Digital:** Uses electronic detectors to record light
- ▶ **Detectors:** Light-sensitive materials at the focal plane
 - ▶ Convert photons into electrical charges



Charge-Coupled Devices (CCD)

- ▶ **CCD Sensors:** Light-sensitive material on a silicon chip
 - ▶ Converts photons into electrical charges
- ▶ **Array of Detectors:**
 - ▶ **Linear arrays:** Single line of detectors
 - ▶ **Two-dimensional arrays:** Multiple rows and columns
- ▶ **Shift Registers:** Move charges from one detector to the next
 - ▶ Information read sequentially
- ▶ **Advantages:**
 - ▶ Compact, efficient at detecting photons
 - ▶ Responds linearly to brightness



CCD vs. CMOS Sensors

► CCD Advantages:

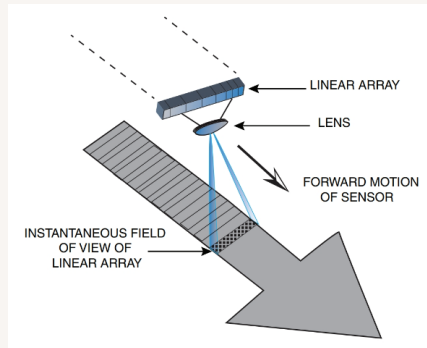
- Efficient in detecting photons, especially in **low light**
- **Linear response** to brightness
- Proven reliability for **scientific imaging**

► CCD Disadvantages:

- Expensive to manufacture
- **High power consumption**
- Requires **analog-to-digital conversion**

► CMOS Sensors:

- Early models had **low sensitivity**, high noise
- Modern CMOS: **Wide dynamic range**, **low power use**
- Increasing use in **aerial cameras**



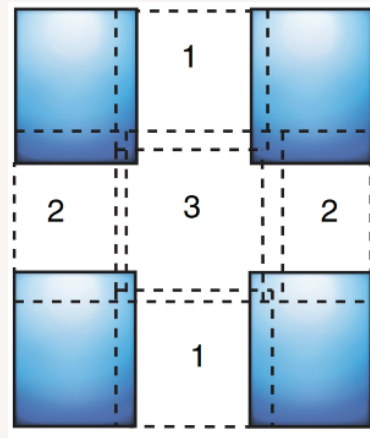
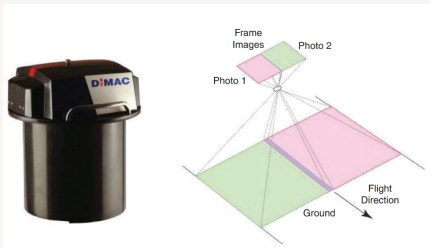
Transition from Analog to Digital Cameras

- ▶ **Analog cameras:** Standard size of **9 in. × 9 in.** frame
 - ▶ Frame-by-frame image capture
 - ▶ Mapping standard: **230 mm × 230 mm**
- ▶ **Digital cameras:** New designs required
 - ▶ No need for traditional formats in digital systems
 - ▶ Larger detector arrays needed for similar coverage
- ▶ **Challenges:** 660-megapixel arrays required to match analog coverage

Solutions for Large-Format Digital Cameras

► Multiple-area CCDs:

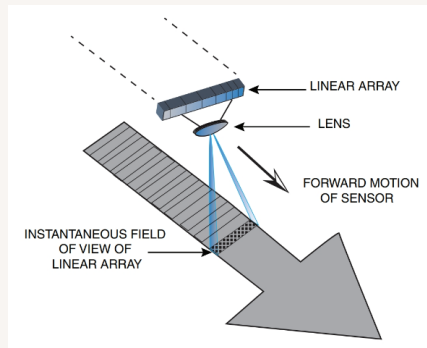
- Use multiple **lens systems** to capture quadrants
- Digitally stitch quadrants into a composite image
- Each lens has its own **nadir** and **brightness adjustments**



Solutions for Large-Format Digital Cameras

► Linear arrays:

- Separate **nadir**, **forward-looking**, and **aft-looking** positions
- Captures only a few lines at a time
- Assembles full imagery as aircraft moves forward



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Digital Scanning of Analog Images

- ▶ **Analog images** are scanned to create **digital versions**
 - ▶ Facilitates **storage, transmission, and analysis**
- ▶ **Office scanners:** Suitable for casual use, but not for scientific applications
 - ▶ Limited positional accuracy
- ▶ **High-quality flatbed scanners:** Required for scientific and photogrammetric work
 - ▶ Large scanning surfaces, **large CCD arrays**, and advanced software
- ▶ **Scanning:** An improvisation compared to **original digital data collection**



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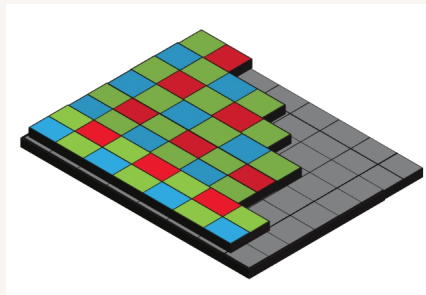
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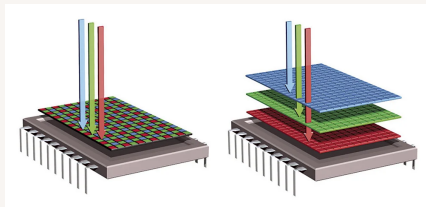
Spectral Sensitivity in Digital Sensors

- ▶ **Spectral sensitivity:** Sensors detect specific regions of the spectrum
 - ▶ Visible spectrum, with a peak in **green**, extends into **near-infrared (NIR)**
- ▶ **Bayer filter:** Allocates pixels to capture **green (50%)**, **red (25%)**, and **blue (25%)**
 - ▶ Filters specific wavelengths reaching each pixel
 - ▶ Interpolation used to estimate missing values for each color
- ▶ **Digital sensors** offer high precision in color brightness



Bayer Filter vs. Foveon Technology

- ▶ **Bayer filter** limitations:
 - ▶ Color filters reduce energy reaching the sensor
 - ▶ **Interpolation** reduces sharpness
- ▶ **Foveon technology**: A more advanced alternative
 - ▶ Three **separate detector layers**: Blue at surface, green below, red deepest
 - ▶ No filters required, each pixel captures all colors
 - ▶ *Kind of exploits color aberation??*
- ▶ Concerns: Deeper layers may receive weaker signals, causing **higher noise levels**



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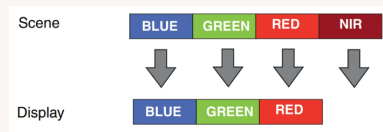
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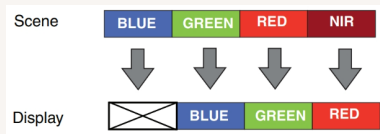
Band Combinations: Optical Imagery

- ▶ **Band combinations:** Assign colors to brightness values in different spectrum regions
- ▶ **Human vision:** Sensitive to three additive primaries: **blue**, **green**, **red**
- ▶ **Digital flexibility:**
 - ▶ Can represent non-visible spectrum regions using colors
- ▶ **Band selection:** Defined by conventions and practical use



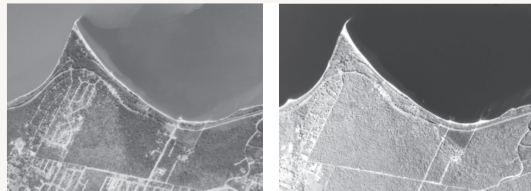
Band Selection for Visible and Non-Visible Spectra

- ▶ **Visible spectrum:**
 - ▶ Blue objects displayed as **blue**
 - ▶ Green objects as **green**, red objects as **red**
- ▶ **Non-visible spectrum:**
 - ▶ Arbitrary color assignments needed
 - ▶ **Near-infrared (NIR):** No fixed primary color, assignment is conventional
- ▶ **Effective band combinations** are built from experience and established practice



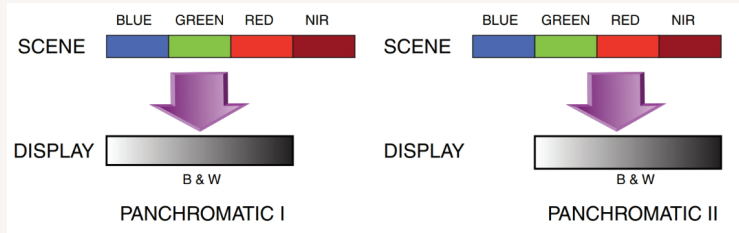
Black-and-White Infrared Imagery

- ▶ **Near-infrared (NIR):** Free from atmospheric scattering
 - ▶ Highlights **vegetation** and **land-water distinctions**
- ▶ **Captured using NIR sensors:**
 - ▶ Filters out visible light
 - ▶ Displays **brightness values** from NIR region
- ▶ Highly valued in **remote sensing**



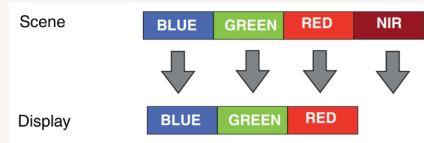
Panchromatic Imagery

- ▶ **Panchromatic:** Represents the **visible spectrum** as a single channel
- ▶ **Black-and-white images:** Record brightness without separating colors
- ▶ Focus on **spatial detail** over color representation
 - ▶ Ideal for **urban imagery**
- ▶ Extended to include **NIR**: Sometimes designated as **VNIR**



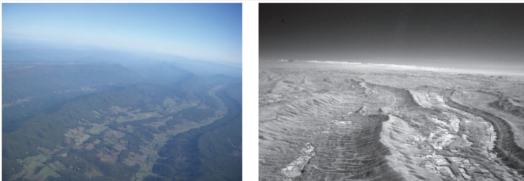
Natural-Color Model

- ▶ **RGB model:** Blue = blue, green = green, red = red
- ▶ **Familiar representation:** Common in **color films** and **displays**
- ▶ **Limitations:** Blue region is affected by **atmospheric scattering**
 - ▶ Reduces clarity at **high altitudes**



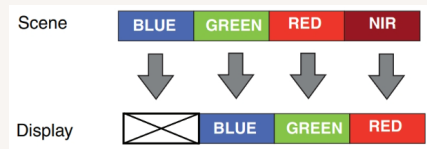
Beyond the Visible Spectrum

- ▶ Remote sensing collects data beyond the **visible spectrum**
 - ▶ False-color images created for non-visible regions
- ▶ **False-color images:** Arbitrary color assignments
 - ▶ Useful for specific objectives, like detecting vegetation or water
- ▶ No “correct” way to assign colors to non-visible data



Color-Infrared (CIR) Model

- ▶ **NIR region:** Adds valuable info on **vegetation**
 - ▶ Not affected by **atmospheric scattering**
- ▶ **CIR model:** Uses three bands
 - ▶ **NIR channel** replaces the **blue band**
- ▶ **Developed in WWII** for **camouflage detection**
- ▶ Excellent for **vegetation** and **water body** detection





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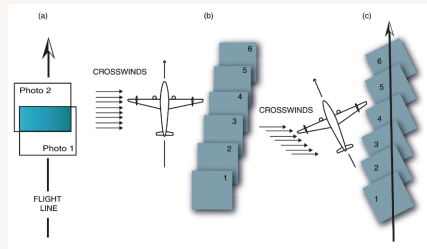
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Coverage by Multiple Photographs

- ▶ **Flight lines:** Parallel strips of photographs
 - ▶ Built for full coverage of an area
- ▶ **Forward overlap:** 50–60% to avoid gaps
- ▶ **Drift:** Deflection of aircraft's course by crosswind
- ▶ **Crab:** Compensation for drift without camera reorientation



Calculating Number of Photos

- ▶ Formula to calculate number of photos:

$$\text{Number of photos} = \frac{\text{Length of flight line}}{(\text{gd of photo}) \times (1 - \text{overlap})}$$

- ▶ **gd**: Ground distance covered by one photo
- ▶ Example:
 - ▶ Flight line: 33 miles
 - ▶ Ground distance: 3.4 miles
 - ▶ Overlap: 60%
 - ▶ **Result**: 25 photos

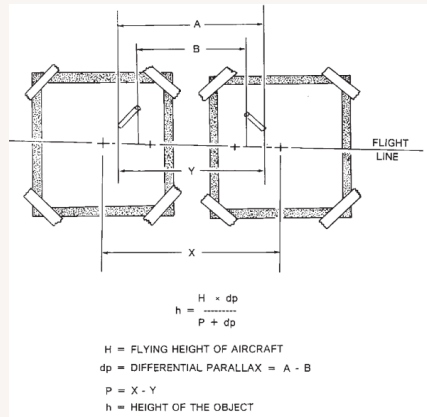
Stereoscopic Parallax

- ▶ **Parallax:** Displacement of objects due to different perspectives
- ▶ Nearby objects show **more parallax**
- ▶ **Overlapping photos** allow for stereoscopic analysis
- ▶ **Parallax measurement:** Basis for estimating object height
- ▶ Displacement always **parallel to flight line**



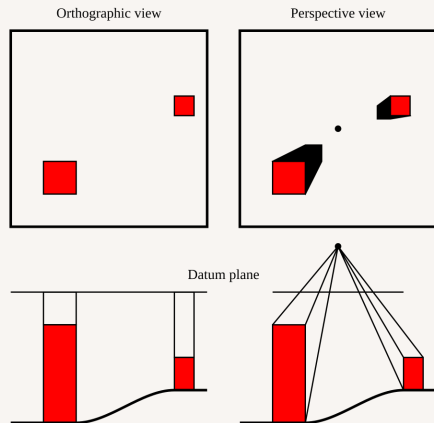
Measuring Parallax

- ▶ **Manual parallax measurement:**
 1. Measure **distance between principal points (X)**
 2. Measure distance between **base images (Y)** and subtract from (X) to get **P**
 3. Measure top-to-top (B) and base-to-base (A) distances, subtract to get **dp**
- ▶ **Parallax shift** is greatest for tall objects close to the camera



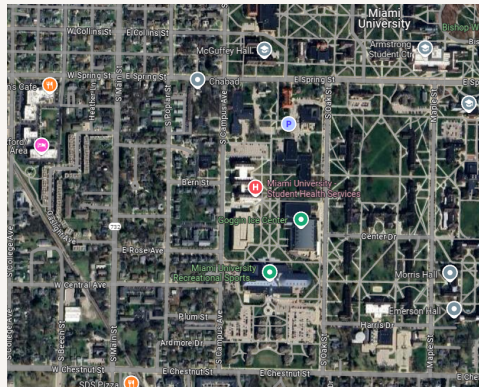
Orthophotos

- ▶ **Orthophotos:** Corrected aerial photographs
- ▶ Remove errors from **tilt and relief displacement**
- ▶ Provide **accurate planimetric positions**
- ▶ Generated using **terrain data** and **stereoscopic images**
- ▶ Used for **GIS and mapping** due to accuracy
- ▶ Umm...
 - ▶ Usually height is estimated at 30m per pixel
 - ▶ (no good for buildings...)



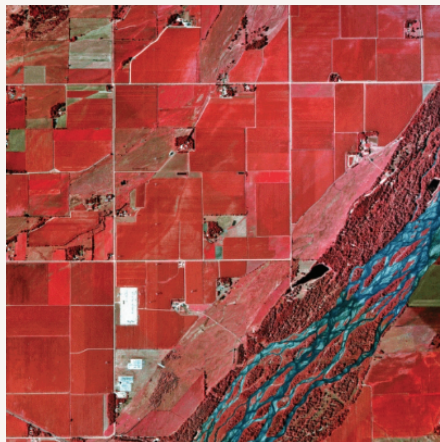
Orthophotomaps

- ▶ **Orthophotomaps:** Combine orthophotos with **map details**
- ▶ Include **place names, symbols, coordinates**
- ▶ Provide **consistent scale** across the image
- ▶ Valuable as **map substitutes** for quick, detailed information
- ▶ Widely used in **GIS** and **topographic mapping**



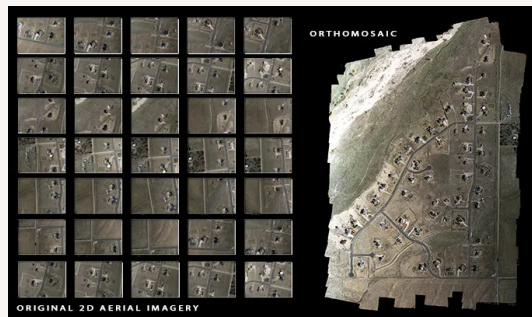
Digital Orthophoto Quadrangles

- ▶ **DOQs:** Digital orthophotos aligned to **USGS 7.5-minute quadrangles**
- ▶ Rectified using **Digital Elevation Models (DEMs)**
- ▶ Available as **black-and-white** or **color images**
- ▶ Provide **2 m image detail** (1 m for DOQQs)
- ▶ Used by **USGS** for **nationwide cartographic data**



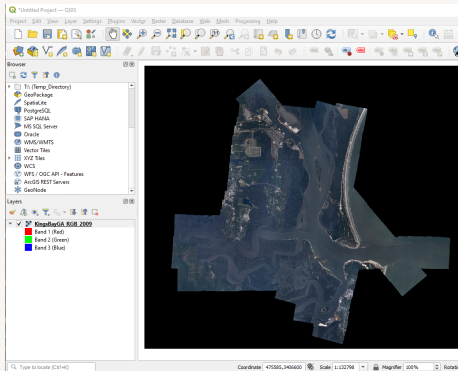
Mosaicking and Matching DOQs

- ▶ **Mosaicking:** Combines adjacent DOQs for seamless coverage
- ▶ **Matching Borders:** Includes 50–300 m border for **image alignment**
- ▶ **Seamless imagery** over large areas
- ▶ Used in **GIS** applications for continuous mapping
- ▶ Supported by **USGS** and **USDA**



VRT Files for Raster Data

- ▶ **VRT (Virtual Raster):** XML-based file for raster datasets
- ▶ **Reference Multiple Rasters** without duplication
- ▶ Used to **mosaic** and **organize** large datasets
- ▶ **Dynamic:** Update raster data without recreating files
- ▶ Ideal for **large-scale remote sensing** projects
- ▶ Works as if a single **HUGE** image in **rastrerio**
 - ▶ Never read the entire thing into RAM at once
 - ▶ Can read blocks that span multiple tiles though!



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4.6 Spectral Sensitivity

4.7 Band Combinations in Optical Imagery

4.8 Coverage by Multiple Photographs

4.9 Photogrammetry

4.10 Acquiring Aerial Photography

Resources

Photogrammetry Overview

- ▶ **Science of measuring from photos**
- ▶ Reconstructs object dimensions and positions
- ▶ Uses knowledge of **camera geometry**
- ▶ Frequently applied to **stereo aerial photography**
- ▶ Key in **topographic mapping**



Modern Photogrammetry

- ▶ **Softcopy photogrammetry:** Digital process
- ▶ Uses GPS and **inertial navigation** data
- ▶ Faster, more accurate than analog methods
- ▶ Produces planimetrically correct images
- ▶ **Multiray photogrammetry:** Higher detail from multiple views



Roadmap

4.1 Introduction

4.2 Fundamentals of the Aerial Photograph

4.3 Geometry of Vertical Aerial Photographs

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Resources

Acquiring Aerial Photography

- ▶ **Custom acquisition:** Contracted aerial firms
- ▶ **Archival imagery:** Available for purchase
- ▶ Archival images may not meet exact specifications but are cost-effective



EROS Data Center

- ▶ Operated by **USGS**
- ▶ Repository for aerial & satellite images
- ▶ Offers a computerized database for easy access
- ▶ Source of imagery from **NASA, USGS, and other agencies**



National Aerial Photography Program (NAPP)

- ▶ Managed by **USGS** since 1987
- ▶ Provides systematic aerial coverage
- ▶ **1:40,000 scale** photography with **60% forward overlap**
- ▶ Primarily **black-and-white** or **color infrared** imagery
- ▶ Used for land surveys, agriculture, and environmental monitoring



National Agricultural Imagery Program (NAIP)

- ▶ Focuses on **agricultural imagery** during growing seasons
- ▶ Provides **digital orthophotography** in color or CIR at **1-m resolution**
- ▶ Primarily used for agriculture but applicable to urban planning and resource assessments



Roadmap

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Resources

Teaching and Learning Resources

- ▶ Additive vs Subtractive Color: YouTube Video
- ▶ CMYK and RGB Color Modes: YouTube Video
- ▶ Analog-to-Digital Mapping: YouTube Video
- ▶ Aerial Survey Photography: YouTube Video
- ▶ Aerial Photography Basics: YouTube Video
- ▶ Bayer Sensor Explained: YouTube Video
- ▶ Camera Lens Magnification: YouTube Video
- ▶ Compact Digital Camera Function: YouTube Video
- ▶ Aero Triangulation: YouTube Video